

Quantifying uncertainty in affect modelling

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Abstract

Recognising human emotional states is a fundamental yet notoriously difficult task in affective computing, largely due to the subjective and ambiguous nature of emotions. This inherent subjectivity directly impacts the reliability of emotion recognition models, highlighting a critical need for methodologies that can accurately assess and report prediction uncertainty. This dissertation investigates the application of the conformal prediction framework to quantify uncertainty in affect modelling. The primary aim of this research is to develop machine learning algorithms that not only recognise human emotions but also provide statistically valid confidence levels for their predictions.

To achieve this, conformal prediction algorithms were implemented and rigorously evaluated against conventional machine learning baselines, including Linear models, Multi-Layer Perceptrons (MLPs), and Long Short-Term Memory (LSTM) networks. The comparative evaluation was conducted using two diverse, publicly available multimodal datasets: the Arousal Video Game Annotation (AGAIN) dataset and the RECOLA collaborative affective interactions corpus. The models were systematically assessed based on both their computational efficiency and recognition accuracy across continuous and discrete affect recognition tasks.

Results indicate that the conformal prediction framework successfully calibrates the models, outputting reliable prediction intervals without significantly compromising standard accuracy metrics or introducing prohibitive computational overhead. Specifically, the conformal predictors achieved valid empirical coverage at the targeted confidence levels. The findings demonstrate that integrating conformal prediction offers a highly successful, model-agnostic approach to handling the subjectivity of human emotions in AI and ML systems. Ultimately, this work contributes a comprehensive evaluation of uncertainty quantification in affective computing and delivers an open-access code repository to facilitate future research in reliable and trustworthy emotion recognition.

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List of Abbreviations

AGAIN	Arousal Video Game Annotation dataset
AI	Artificial Intelligence
AM	Affect Modelling
AVEC	Audio-Visual Emotion Challenge
BNN	Bayesian Neural Network
CCC	Concordance Correlation Coefficient
ComParE	Computational Paralinguistics Challenge feature set
CP	Conformal Prediction
ERC	Emotion Recognition in Conversation
FYP	Final Year Project
ICP	Inductive Conformal Prediction
LSTM	Long Short-Term Memory network
MFCC	Mel-Frequency Cepstral Coefficient
MLP	Multi-Layer Perceptron
MPIW	Mean Prediction Interval Width
MPSS	Mean Prediction Set Size
MSE	Mean Squared Error
RECOLA	Remote Collaborative and Affective Interactions corpus
ReLU	Rectified Linear Unit
RMSE	Root Mean Squared Error
UQ	Uncertainty Quantification

1 Introduction

Affective computing is a discipline concerned with developing systems capable of recognising, interpreting, and responding to human emotional states [1], [2]. Over the past two decades, significant advances in deep learning have substantially improved the accuracy of emotion recognition models across a range of modalities, including audio, video, and physiological signals [3], [4]. Despite these improvements, a fundamental limitation persists: the vast majority of current models produce a single point prediction, a scalar value representing the most likely emotion estimate, with no accompanying measure of confidence or uncertainty. In safety-critical and human-facing applications, this limitation is not merely a theoretical inconvenience. A model that is unable to communicate when it is uncertain is a model that cannot be trusted.

This dissertation investigates the application of Conformal Prediction (CP) [5], [8] as a principled framework for quantifying uncertainty in affect modelling. Conformal prediction is a distribution-free, model-agnostic technique that wraps around any pre-trained machine learning model and converts its scalar outputs into statistically valid prediction intervals for regression tasks or prediction sets for classification tasks, at a user-specified confidence level. Critically, these guarantees hold under the mild assumption of exchangeability, without requiring any assumptions about the underlying data distribution. The framework is evaluated on two publicly available affective datasets, the RECOLA multimodal corpus [1] and the AGAIN arousal annotation dataset [2], across three baseline model architectures of increasing complexity.

1.1 Problem Definition

The core problem addressed in this dissertation arises from the intersection of two well-documented issues in affective computing. The first is the inherently subjective nature of human emotion. Unlike object recognition, where a label such as "this is a car" is objectively verifiable, emotion annotation relies on the aggregated opinions of multiple human annotators who frequently disagree. The RECOLA corpus [1], for instance, captures spontaneous collaborative interactions in which trained

annotators nonetheless exhibit imperfect internal agreement on arousal and valence values over time. Similarly, the AGAIN dataset [2] demonstrates that even within a single gaming genre, individual players exhibit substantially different affective responses to identical game events. The standard practice of aggregating these annotations into a single mean value, often referred to as the "gold standard", effectively discards the disagreement signal entirely. This unavoidable noise, which arises from the data itself rather than the model, is formally referred to as aleatoric uncertainty [5], [13].

The second issue is that conventional machine learning approaches do not model this uncertainty in any meaningful way. Models such as linear regressors, MLPs and LSTM networks are trained to minimise a loss function against the mean annotation, and at inference time they output a single scalar prediction. Two critical limitations follow from this approach. First, the prediction provides no information about model confidence: a predicted arousal value of 0.65 communicates nothing about whether the model is highly certain or completely guessing. Second, deep learning models are well known to exhibit overconfidence, frequently assigning high confidence to incorrect predictions when the input is ambiguous or out-of-distribution, both of which are common in large affective datasets.

Taken together, these limitations mean that current affect recognition systems are unable to distinguish between predictions made on clearly expressive samples, represented by highly-discriminative feature values, and predictions on ambiguous samples, where emotional content is genuinely uncertain. Without a rigorous framework to communicate this distinction, deploying such models in real-world adaptive systems, such as games that modify difficulty based on player arousal, or healthcare tools that monitor patient distress, introduces unquantified risk. If a game responds to a low arousal prediction that the model was in fact highly uncertain about, the result may be a frustrating and counterproductive experience for the user.

1.2 Motivation

The primary motivation for this work is the disconnect between the subjective, probabilistic nature of human emotion and the deterministic, overconfident outputs of current affective computing systems. As the RECOLA and AGAIN datasets both

illustrate, human emotional experience is not a single ground truth value but rather a distribution of subjective interpretations. A computational system that ignores this distributional character and produces a point estimate is, in a practical sense, making a stronger claim about emotional state than the underlying evidence supports.

Existing approaches to uncertainty quantification in deep learning, such as Bayesian Neural Networks and Monte Carlo Dropout [3], partially address this issue but come with significant practical drawbacks. Bayesian methods require modifying the model architecture and training procedure, and their resulting uncertainty estimates rely on prior assumptions that may not hold for real affective data. Monte Carlo Dropout increases inference latency by a factor equal to the number of forward passes, which is problematic in real-time applications such as interactive gaming. Deep Ensembles [4] produce strong uncertainty estimates but multiply training cost linearly with ensemble size. Most critically, none of these approaches provides a formal statistical guarantee on the coverage of their uncertainty estimates, they can still be "confidently wrong".

Conformal prediction addresses precisely this gap. It requires no modification to the base model architecture or training procedure, it introduces negligible computational overhead at inference time, and it provides a mathematically proven marginal coverage guarantee that does not depend on distributional assumptions [5]. In the context of affective computing, this means that a conformal regressor can wrap any pre-trained emotion recognition model and guarantee, for example, that the true arousal value falls within the predicted interval at least 90% of the time on unseen participants. This kind of honest uncertainty communication is a prerequisite for deploying affect-aware systems in contexts where user trust and model transparency matter, from adaptive entertainment to clinical monitoring.

An additional motivation is the relative absence of conformal prediction applications in affective computing. While CP has seen growing adoption in domains such as medical diagnosis and natural language processing, its application to high-frequency, continuous arousal and valence signals, particularly in gaming contexts, remains largely unexplored. Recent work by Li et al. [6] applied CP to depression severity scores derived from facial features, and demonstrated its utility in emotion recognition from conversation. However, neither addresses the specific challenges

posed by continuous affective time-series in interactive environments, where non-stationarity and inter-subject variability are significant confounding factors. This dissertation targets this gap directly.

1.3 Aims and Objectives

The primary aim of this dissertation is to investigate the applicability of the conformal prediction framework for quantifying uncertainty in affective computing tasks. Specifically, the project seeks to determine whether conformal predictors can provide statistically valid confidence guarantees for continuous and discrete emotion recognition without significantly sacrificing model performance or computational tractability. To fulfil this aim, the following five technical objectives were defined at the outset of the project and are referred to throughout the remainder of this dissertation:

1. **Data Acquisition and Preprocessing:** To acquire and preprocess the RECOLA and AGAIN datasets for use in both regression and classification affect recognition tasks. Preprocessing involves standardising input features using participant-independent normalisation, constructing temporal sliding windows to provide sequential context to the models, and discretising continuous affect labels into five balanced ordinal classes using training-set-derived percentile boundaries.
2. **Development of Baseline Models:** To implement three baseline model architectures of increasing complexity: a Linear model, an MLP, and an LSTM network. Each architecture is applied to both regression and classification formulations of the arousal and valence prediction tasks across both datasets, providing a comprehensive performance benchmark.
3. **Implementation of Conformal Predictors:** To wrap each trained baseline model with an Inductive Conformal Predictor using the crepes library [8], producing prediction intervals for regression tasks and prediction sets for classification tasks. Conformal calibration is performed on a held-out participant group to preserve subject independence, and predictors are evaluated at five significance levels corresponding to target confidence levels of 99%, 95%, 90%, 80%, and 60%.
4. **Evaluation Protocol:** To evaluate all models using a participant-independent train-validation-test split strategy based on GroupShuffleSplit, ensuring that no

participant appears in more than one split. This design prevents data leakage across subjects and provides a realistic estimate of generalisation to unseen individuals, which is the most relevant deployment scenario for affect-aware systems.

5. **Comparative Analysis:** To benchmark the conformal predictors against the baseline models using a dual-metric evaluation framework. Predictive accuracy is assessed using CCC and RMSE for regression, and accuracy for classification. Uncertainty quantification is assessed using empirical coverage rate and mean prediction interval width for regression, and empirical coverage rate and mean prediction set size for classification. The analysis specifically examines whether the theoretical coverage guarantee is upheld and how interval efficiency varies across model architectures and datasets.

1.4 Document Structure

The remainder of this dissertation is structured as follows. Chapter 2 presents the technical background, covering affective computing, audio feature extraction, the baseline architectures, the aleatoric–epistemic distinction, and the foundations of conformal prediction. Chapter 3 reviews the related literature on affect modelling, uncertainty quantification in deep learning, and prior applications of conformal prediction, culminating in the research gap addressed by this work. Chapter 4 sets out the specification and design, detailing the pipeline architecture, dataset characteristics, preprocessing decisions, and the baseline and conformal design choices, with explicit justification. Chapter 5 covers the implementation, describing the software components developed and the technical decisions taken during development. Chapter 6 presents the testing and evaluation, reporting baseline performance and the full conformal evaluation across all models, datasets, and confidence levels. Chapter 7 discusses future work, and Chapter 8 concludes the dissertation with a summary of contributions and reflections on the research.

2 Background

This chapter establishes the technical foundations required to understand the research presented in this dissertation. It is organised into five areas: the fundamentals of affective computing and emotion modelling, the audio feature representations used for affect recognition, the machine learning architectures employed as baselines, the problem of uncertainty in deep learning systems, and the mathematical principles underpinning the conformal prediction framework.

2.1 Affective Computing and Emotion Modelling

Affective computing is the field concerned with enabling machines to recognise, interpret, and simulate human emotional states. The term was introduced by Picard [14], who argued that machines capable of responding appropriately to human affect would be fundamentally more effective in human-computer interaction. Since then, the field has grown substantially, driven by advances in machine learning and the increasing availability of annotated affective corpora [1], [2], [3].

The foundational question in affective computing is how to represent emotion computationally. Early approaches adopted categorical models, representing emotion as membership in one of a small set of discrete classes such as happiness, sadness, anger, fear, disgust, and surprise, following the universal-emotions taxonomy of Ekman [15]. While intuitive, these categorical representations are limited in their ability to capture the continuous and gradual transitions between emotional states that characterise real human experience. A person does not suddenly switch from neutral to happy, emotion evolves progressively over time.

In response to these limitations, dimensional models of emotion were proposed as a more expressive alternative. The most influential of these is Russell's circumplex model of affect [16], which represents emotional states not as discrete categories but as points in a continuous two-dimensional space defined by two orthogonal axes: arousal, ranging from low activation (calm, bored) to high activation (excited, tense), and valence, ranging from unpleasant to pleasant. Under this framework, any emotional state can be characterised by its coordinates along these two axes, enabling

the representation of subtle, mixed, and transitional affective states that categorical models cannot capture. Crucially, the dimensional model also accommodates the continuous temporal evolution of emotion: as a person's affective state shifts gradually over time, its trajectory can be traced as a smooth path through the arousal-valence plane rather than as a sequence of discrete label transitions. This property makes the dimensional model particularly well suited to time-series annotation tasks such as those employed by RECOLA and AGAIN, where annotators continuously track a subject's arousal or valence over the course of an interaction. Russell's circumplex model is illustrated in Figure 1.

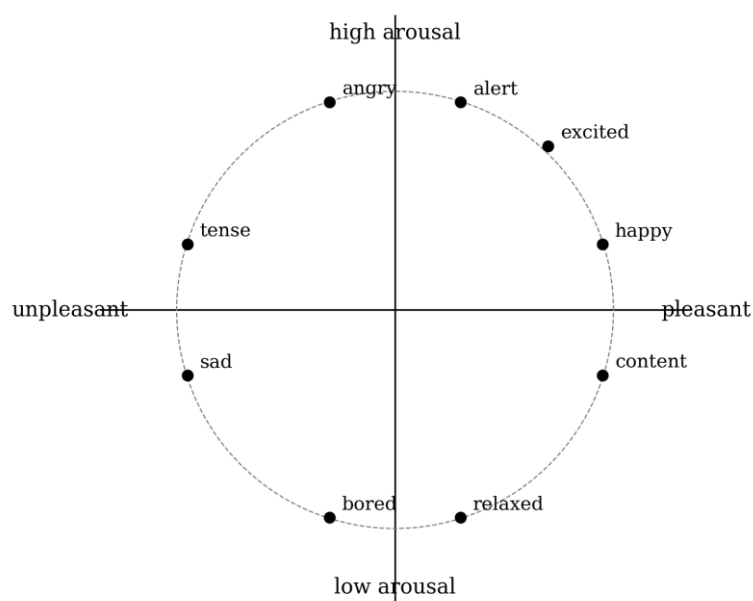


Figure 1: Russell's Circumplex Model of Affect [16]

A persistent challenge in the dimensional model is annotation. Human annotators are asked to continuously rate the arousal and valence of a human user, typically using a joystick or slider interface. It is important to stress that the annotators do not annotate the stimulus itself, rather, they infer and label the emotional state of the human user being observed. Because emotional perception is inherently subjective, different annotators frequently disagree on the precise values. The standard practice is to aggregate these annotations into a single mean trace, which serves as the regression target during model training [4], [17]. This aggregation,

however, discards the variance in annotator opinion, effectively treating a genuinely uncertain emotional signal as if it were a single deterministic ground truth.

2.2 Machine Learning Architectures for Affect Modelling

This section describes the three classes of model architecture used as baselines in this dissertation. Each represents a different level of modelling capacity and inductive bias, ranging from a simple linear model to a temporal neural network capable of learning dependencies across time.

The three baseline architectures are: a Linear model that applies a single affine projection to the final frame of the input window, providing an interpretable lower bound on task difficulty, a Multi-Layer Perceptron (MLP) with two hidden layers (128 and 64 units, ReLU activations, and dropout), enabling higher-order feature interactions, and an LSTM [19] that processes the full 20-frame window through a 64-dimensional recurrent hidden state, capturing temporal dynamics in the affect signal. Full architectural hyperparameters and design rationale are given in Section 4.4.

Model performance in regression tasks is assessed using the Concordance Correlation Coefficient (CCC), defined as:

$$\rho\rho_c = \frac{2\sigma_{y\hat{y}}}{\sigma_y^2 + \sigma_{\hat{y}}^2 + (\mu_y - \mu_{\hat{y}})^2} \quad (2.1)$$

where y denotes the ground truth annotations, $\hat{y}$ the model predictions, $\sigma_{y\hat{y}}$ denotes the covariance between y and $\hat{y}$, σ_y^2 and $\sigma_{\hat{y}}^2$ denote their respective variances, and μ_y and $\mu_{\hat{y}}$ denote their respective means. The CCC ranges from -1 to $+1$, where $+1$ indicates perfect agreement and 0 indicates no relationship. Crucially, unlike Pearson correlation, the CCC penalises both deviations in correlation and in the mean and scale of the prediction [20]. A model that correctly tracks the direction of emotional change but systematically over- or under-predicts absolute values will score lower on CCC than on Pearson correlation. This makes CCC the standard evaluation metric in the AVEC benchmark challenges and in the broader affect modelling literature.

The Root Mean Squared Error (RMSE), defined as the square root of the mean squared difference between predictions and ground truth values, is reported as a complementary regression metric on the same scale as the labels themselves, providing an interpretable measure of absolute prediction error that is directly comparable across model architectures.

Model performance in classification tasks is assessed using top-1 accuracy, defined as the fraction of test examples for which the predicted class matches the true class. Because all classification tasks in this dissertation use five balanced ordinal classes derived from training-set quantiles, a uniform-random baseline yields an expected accuracy of 20%, providing a natural reference point against which the trained classifiers can be compared.

2.3 Uncertainty in Machine Learning

Uncertainty in machine learning systems is broadly categorised into two qualitatively distinct types, each arising from a different source and carrying different implications for how it may, or may not, be reduced (see Figure 2). Aleatoric uncertainty, also referred to as data or irreducible uncertainty, is inherent to the data-generating process itself [13]. It arises when the observed outputs are intrinsically variable given the inputs, for example when multiple annotators assign different arousal ratings to the same stimulus because emotional perception is genuinely subjective. Because this variability reflects a property of reality rather than a deficiency of the model or the dataset, aleatoric uncertainty cannot be reduced by collecting more data or by increasing model complexity; it constitutes a fundamental lower bound on the achievable prediction error. Epistemic uncertainty, by contrast, is attributable to the model rather than the data [13]. It arises from a lack of knowledge, either about the true functional relationship between inputs and outputs, or about regions of the input space that are sparsely represented in the training set. Because epistemic uncertainty reflects ignorance rather than irreducible noise, it is in principle reducible: a model trained on more data or equipped with greater expressive capacity can, in favourable circumstances, achieve lower epistemic uncertainty on previously unseen inputs. In practice, the two types of uncertainty co-exist and are difficult to disentangle. The standard approach of aggregating annotator disagreement into a

single mean label, adopted by both RECOLA and AGAIN, suppresses the visible signal of aleatoric uncertainty and presents the model with a target that conflates both sources.

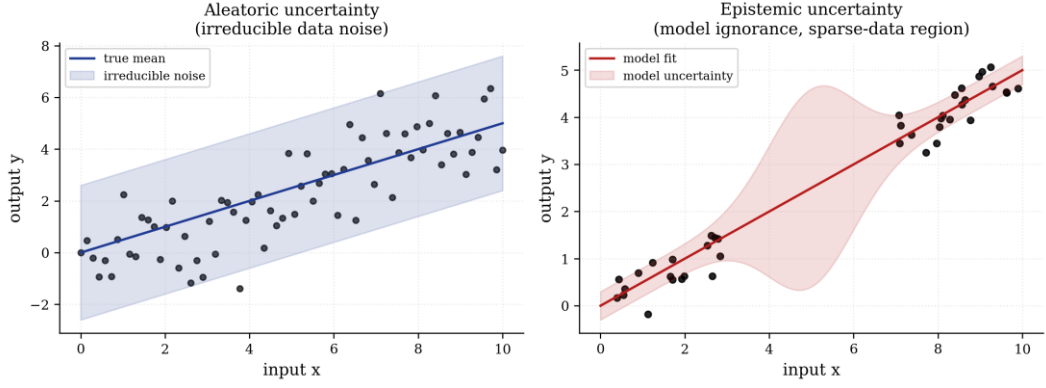


Figure 2: Aleatoric versus epistemic uncertainty

Conventional deep learning models, including those used in this dissertation, are trained to minimise a point-estimate loss such as mean squared error or cross-entropy. At inference time, they produce a single output value with no associated measure of confidence. This is problematic for two reasons. First, the model cannot signal when it is operating outside its training distribution. Second, deep neural networks are well documented to be overconfident, assigning high confidence to incorrect predictions particularly under distribution shift [21]. In safety-sensitive or user-facing applications, this silent overconfidence is a fundamental reliability risk.

Several approaches address this limitation. Bayesian Neural Networks (BNNs) [22] place prior distributions over network weights and compute a posterior at training time, providing principled uncertainty estimates at the cost of intractable exact inference. Monte Carlo Dropout [6] approximates Bayesian inference by retaining dropout at inference time across multiple stochastic forward passes, while computationally inexpensive, the resulting estimates can be poorly calibrated [21]. Deep Ensembles [7] aggregate predictions from multiple independently trained networks and have strong empirical performance, but their cost scales linearly with ensemble size. All three methods require architectural modifications or repeated forward passes, making them impractical in some deployment settings. Conformal

prediction, introduced in Section 2.4, sidesteps these limitations entirely by operating as a post-hoc wrapper around any trained model.

2.4 Conformal Prediction

Conformal prediction is a framework for constructing prediction regions with guaranteed coverage, developed within the framework of algorithmic learning theory by Vovk, Gammerman, and Shafer [8]. Unlike the Bayesian and ensemble-based approaches described in Section 2.3, conformal prediction makes no assumptions about the form of the underlying data distribution and requires no modification to the base model architecture, that is, the machine learning model used to address the problem at hand. Its only theoretical requirement is that the calibration and test data are exchangeable, a condition that is satisfied when data points are drawn independently from the same distribution.

Important definitions to conformal prediction are two concepts that recur throughout this dissertation. The calibration set is a held-out portion of labelled data that is not used during model training. Once a base model has been fitted, it is applied to the calibration set to generate predictions, and the discrepancies between those predictions and the true labels are used to characterise the model's typical error behaviour on unseen data. The non-conformity score is the numerical measure of that discrepancy for each calibration example, quantifying how poorly a given prediction conforms to its observed label. For regression tasks, a natural non-conformity score is the absolute residual. For classification tasks, a common choice is one minus the softmax probability assigned to the true class. The distribution of non-conformity scores across the calibration set encodes the uncertainty of the model in a distribution-free manner: at test time, a quantile of this distribution is used as a threshold to construct the prediction interval or prediction set at the desired confidence level, thereby providing the statistical coverage guarantee that distinguishes conformal prediction from conventional point-estimate methods.

3 Literature Review

This chapter critically reviews the existing body of work across the three domains central to this dissertation: affect modelling and affective datasets, uncertainty quantification in deep learning, and conformal prediction. For each area, the relevant contributions are analysed and their limitations identified, culminating in the articulation of the specific research gap that motivates the present work.

3.1 Affect Modelling and Affective Datasets

A foundational requirement for any data-driven approach to affective computing is the availability of annotated multimodal corpora captured under ecologically valid conditions. Two datasets are central to this dissertation, and their design choices and limitations directly shape the methodological decisions made in this work.

The RECOLA (Remote Collaborative and Affective Interactions) corpus, introduced by Ringeval et al. [1], was collected to capture naturalistic spontaneous affect during a remote collaborative task. Pairs of participants communicated via video link while attempting to reach consensus on a survival scenario. The resulting audio, video, and physiological streams were annotated continuously by trained annotators using the arousal-valence circumplex. A key design feature of RECOLA is that affect is elicited spontaneously rather than performed, producing more nuanced emotional signals than acted datasets. However, this naturalistic quality also introduces substantial inter-annotator disagreement, particularly for valence. The corpus has since become a standard benchmark in the AVEC challenge series.

The AGAIN (Arousal Video Game Annotation) dataset, introduced by Melhart et al. [2], addresses a distinct and more challenging variant of affective annotation: continuous self-reported arousal during interactive gameplay. Unlike passive media consumption, gameplay produces affective states that are deeply dependent on individual skill, play style, and familiarity with the specific game. The AGAIN dataset collected arousal annotations from 124 participants across nine game titles spanning three genres, including Shooter, Platformer, and Racing. Melhart et al. demonstrate

that general-purpose arousal representations are difficult to learn from gameplay data because player responses to identical game events vary considerably. This inter-subject variability is a key source of uncertainty in any model trained on the dataset. The present work focuses specifically on the Shooter genre, which offers the largest single-genre participant pool, comprising 121 players and approximately 162,500 data points, providing sufficient volume for participant-independent evaluation.

Makantasis et al. [3] provides the most directly relevant prior work on predictive modelling using both RECOLA and AGAIN. Their study examined general-purpose feature representations derived from pixels and audio for arousal prediction in gaming contexts, establishing competitive baselines using convolutional and recurrent architectures. The results demonstrate that arousal from audio features in RECOLA achieves a CCC of approximately 0.40 to 0.45 with well-tuned models, while gaming arousal in AGAIN is generally harder to predict, consistent with the higher inter-subject variability in that dataset. These published baselines provide a reference frame for evaluating the models developed in this dissertation.

A broader theoretical concern raised by RECOLA's annotation methodology is the treatment of inter-annotator disagreement. Aroyo and Welty [4] argue that aggregating multiple annotations into a single mean label destroys information: when annotators are genuinely divided, the mean may correspond to a value no individual annotator assigned, producing a fictitious ground truth. This dissertation treats such disagreement as an intrinsic property of the affective signal and uses conformal prediction to propagate it into model output rather than suppress it.

3.2 Conformal Prediction: Theory and Applications

Conformal prediction was introduced by Vovk, Gammerman, and Shafer [8] as a framework for constructing prediction regions with provable coverage guarantees under the mild assumption of exchangeability. Exchangeability is a weaker condition than i.i.d.: it requires only that the joint distribution of calibration and test points is invariant under permutation, which is automatically satisfied when data are drawn i.i.d. from the same distribution. Unlike Bayesian methods, conformal prediction is entirely distribution-free, making no assumptions about the functional form of the data-generating process, the noise distribution, or the model architecture. The

theoretical guarantee is stated as a finite-sample marginal coverage bound. For a significance level $\alpha \in (0, 1)$ and a calibration set of size n , the conformal predictor Γ^α satisfies Equation (2.2):

$$P(Y^{n+1} \in \Gamma^\alpha(X^{n+1})) \geq 1 - \alpha \quad (2.2)$$

where the probability is over the joint draw of the calibration and test points. Shafer and Vovk further show that the bound is tight in the sense that the coverage cannot significantly exceed $1 - \alpha$ without the intervals becoming unnecessarily wide, provided the calibration scores are distinct. This tightness property is important because it ensures that the predictor is not simply conservative but is as informative as the calibration data permits.

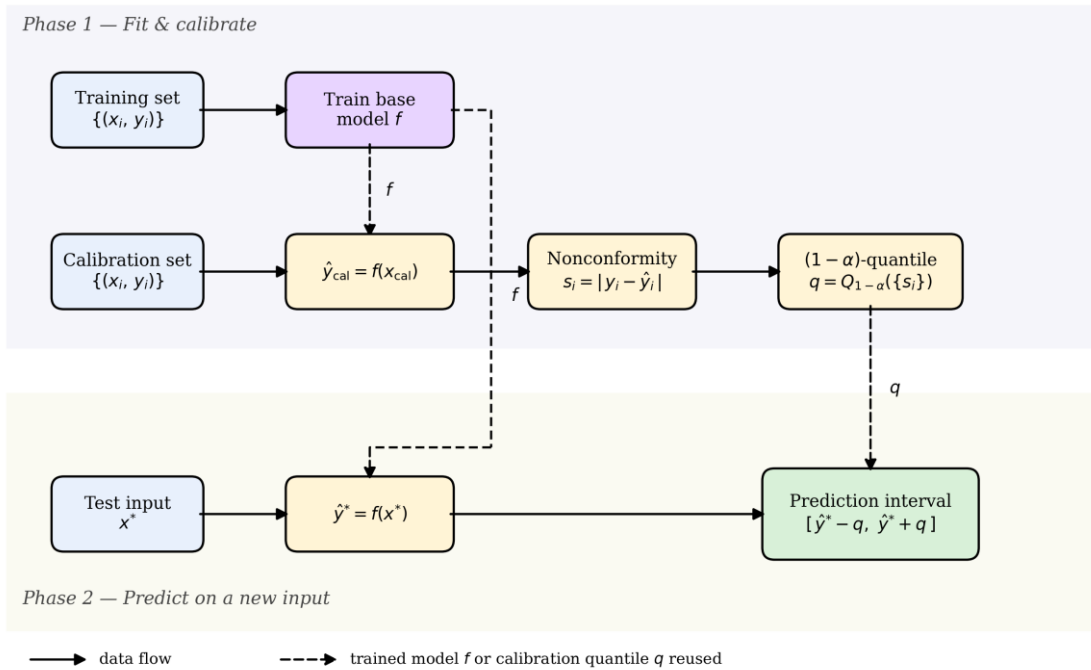


Figure 3: Inductive Conformal Prediction pipeline

Molnar [9] provides a practical introduction to conformal prediction with Python-based implementations, demonstrating the framework across regression and classification tasks with real datasets. The coverage guarantee is illustrated empirically alongside its theoretical derivation, and the key trade-off between coverage validity

and interval efficiency is discussed in depth. A central practical insight from Molnar’s treatment is that the efficiency of a conformal predictor, measured by mean interval width for regression or mean set size for classification, is directly bounded by the accuracy of the underlying model. A model with low residuals produces a tight calibration distribution and hence narrow intervals, while a poorly calibrated model produces wide, less informative intervals. This relationship motivates the evaluation strategy used in Chapter 6, which examines efficiency across model architectures of increasing predictive power.

The *crepes* library, documented by Boström [10], provides a modular and well-tested Python implementation of Inductive Conformal Prediction for both regression and classification. The library implements split conformal predictors alongside Mondrian variants for conditional coverage and difficulty-based normalisation for adaptive intervals. A notable design decision is the separation of non-conformity score computation from calibration and prediction steps, making it straightforward to apply different non-conformity functions to the same base model. This modularity is particularly relevant here, where the same three architectures are evaluated under both regression and classification CP wrappers.

Recent applications of conformal prediction in affective computing and adjacent fields demonstrate both the promise and the limitations of the framework. Li et al. [11] applied CP to facial depression severity analysis, finding that conformal intervals effectively captured the ambiguity in clinical depression scores that standard regressors reported with false precision. Their work showed that CP intervals were systematically wider for participants with borderline clinical assessments, indicating that the non-conformity scores were tracking genuine label ambiguity rather than simply inflating all intervals uniformly. This finding directly parallels the annotation disagreement problem in RECOLA, where ambiguous affective stimuli produce high annotator variance that the model should ideally communicate through wider uncertainty bounds. AGAIN does not exhibit the same disagreement problem, since each data point carries a single self-reported arousal value, but it does exhibit closely related inter-participant differences whose effect on the conformal intervals is examined empirically in Chapter 6.

S. Roohi et al. [12] applied conformal prediction to Emotion Recognition in Conversation (ERC), a task characterised by inherent label ambiguity and context-dependence. Their framework produced dynamic prediction sets that included multiple plausible emotion labels for ambiguous dialogue turns, for example predicting the set {Happy, Neutral} rather than a single label when the dialogue was contextually ambiguous. The authors demonstrated that coverage guarantees were maintained across multiple emotion categories and that set sizes increased predictably as confidence requirements were raised. This work establishes a direct precedent for applying CP to classification-formulated emotion recognition, as done in the classification experiments of this dissertation.

Despite these contributions, a clear gap remains. Neither Li et al. nor S. Roohi et al. address the specific challenges of continuous, high-frequency affective time-series in interactive contexts. Depression severity scores are static assessments, conversational emotion labels are assigned at the utterance level, typically one to several seconds in duration. The AGAIN dataset, by contrast, requires affect prediction at 25 Hz across sessions lasting several minutes, with pronounced temporal autocorrelation and substantial inter-session non-stationarity due to varying gameplay events. Under these conditions, the exchangeability assumption that underpins the conformal coverage guarantee is more likely to be strained, and the empirical coverage may fall below the nominal level for some model and dataset combinations. The Inductive Conformal Prediction pipeline (Figure 3) was developed and used to investigate this question directly, treating any observed coverage shortfall not as a failure of the implementation but as an empirical finding about the conditions under which CP guarantees hold in affective time-series.

3.3 Research Gap and Positioning

The review above reveals a consistent pattern across the literature. Affective computing research has produced increasingly accurate predictive models for arousal and valence, including competitive baselines on RECOLA and AGAIN. Uncertainty quantification research has produced theoretically principled methods for communicating predictive confidence. Yet these two lines of work have not been brought together in a way that provides formal, finite-sample coverage guarantees for

continuous affect recognition, particularly in interactive scenarios where the underlying data are temporally correlated and inter-subject variability is large. The specific contributions of this dissertation in relation to the literature are threefold.

First, to the best of the author’s knowledge, this is the first work to apply Inductive Conformal Prediction systematically to both arousal regression on the AGAIN gaming dataset and to arousal and valence regression on the RECOLA corpus within a unified, participant-independent pipeline. While prior work has applied CP variants to depression severity [11] and to conversational emotion classification [12], no published study, to the author’s knowledge, has reported empirical coverage and efficiency for high-frequency arousal regression on either RECOLA or AGAIN under participant-independent splits. Second, this work provides a systematic comparative evaluation of CP across three model architectures and two datasets, enabling conclusions about how architecture complexity interacts with conformal efficiency.

Also, it examines the empirical behaviour of the coverage guarantee under the temporal autocorrelation conditions characteristic of interactive affective tasks. More generally, any affect corpus representing an interactive task, not only gaming, exhibits temporal correlations that may strain the exchangeability assumption, and the present work provides evidence about whether the assumption holds in practice for this broader application domain.

These contributions are directly tied to the five objectives defined in Section 1.3 and are evaluated against them in Chapter 6. The methodological decisions described in Chapter 4, from the choice of participant-independent splits for both datasets to the selection of the Shooter genre for AGAIN and the window size, are grounded in the literature reviewed above and are justified explicitly in the following chapter.

4 Specification and Design

This chapter outlines the design decisions and the overall system architecture underpinning the experimental pipeline. The pipeline is composed of six sequential stages: data acquisition and inspection, preprocessing (including participant-independent splitting, feature standardisation, temporal windowing, and label discretisation), baseline model training, conformal calibration, evaluation, and result aggregation. Each stage is encapsulated in a dedicated Python module, ensuring reproducibility, modularity, and a clear separation of concerns. The remainder of this chapter discusses each stage in turn, providing both the rationale for the design choices and the technical specifications that the implementation in Chapter 5 instantiates.

4.1 System Architecture Overview

The pipeline is designed to be dataset-agnostic at the training and conformal calibration stages, so that the same model classes and conformal wrappers apply to both RECOLA and AGAIN without modification. Dataset-specific logic—file parsing, feature-column selection, regression-target choice, and the genre filter for AGAIN—is isolated entirely within the preprocessing module. All intermediate outputs are persisted to disk: preprocessed arrays as NumPy compressed files (.npy), trained weights as PyTorch state dictionaries (.pth), and evaluation results as plain-text log files, so that any stage can be re-run independently and the full set of experiments is reproducible.

The architecture is best understood as a directed acyclic graph in which raw data flows from disk through inspection and preprocessing into model-ready tensors, those tensors flow into the training stage to produce trained model checkpoints, and the checkpoints together with held-out calibration and test arrays flow into the conformal stage to produce empirical coverage and efficiency metrics. There are no cyclic dependencies between stages, and the only shared state is the on-disk artefact directory. This permits the conformal stage to be re-evaluated under different

significance levels, or with different non-conformity scores, without any impact on the upstream training or preprocessing logic.

4.2 Dataset Specification

Two datasets are used in this dissertation. Their selection was motivated by the desire to evaluate conformal prediction across two qualitatively different affective tasks: spontaneous dyadic interaction captured in a controlled laboratory setting, and self-reported arousal collected from individual players during interactive gameplay. The RECOLA corpus provides synchronised audio-visual recordings of 46 French-speaking participants engaged in a remote collaborative survival task, together with continuous arousal and valence annotations from six trained annotators sampled at 25 Hz. For this work, the audio modality is used exclusively, represented by the 130-dimensional ComParE 2013 acoustic feature set [18] distributed with the corpus, comprising 65 low-level descriptors. The regression targets are the per-frame median arousal and median valence values derived from the six annotator traces, following the convention of the AVEC challenge series. RECOLA therefore yields four task variants: arousal regression, arousal classification, valence regression, and valence classification.

The AGAIN dataset provides self-reported arousal annotations from 124 participants across nine game titles spanning three genres: Shooter, Platformer, and Racing. Each player annotates their own arousal continuously during gameplay, producing a single arousal trace per session at a frame-level sampling rate. The Shooter genre is selected for this work for two reasons. First, it offers the largest single-genre participant pool, comprising 121 unique players and approximately 162,500 data points after the genre filter is applied, which is more than sufficient for participant-independent evaluation. Second, restricting attention to a single genre controls for genre-level confounds in arousal dynamics that have been documented by Melhart et al. [2], mixing genres would confound the inter-subject variability of interest with the inter-genre differences in event structure. Because AGAIN provides only a single regression target, the dataset yields two task variants: arousal regression and arousal classification.

Across both datasets, then, the experimental matrix consists of six (dataset, target, task) combinations: RECOLA arousal regression, RECOLA arousal classification, RECOLA valence regression, RECOLA valence classification, AGAIN arousal regression, and AGAIN arousal classification. Each combination is independently fit and conformally calibrated for each of the three baseline architectures described in Section 4.4, producing a total of eighteen experimental runs reported in the evaluation chapter.

4.3 Preprocessing Design

Preprocessing is the most consequential stage of the pipeline because, as discussed in Section 2.4, the validity of the conformal coverage guarantee depends critically on the exchangeability of the calibration and test data, and on the absence of any leakage between the training, calibration, and test sets. The preprocessing design therefore proceeds through four carefully ordered steps: participant-independent splitting, feature standardisation, temporal windowing, and label discretisation.

4.3.1 Participant-Independent Splitting

For affective computing tasks, the most realistic and demanding deployment scenario is that the system encounters a new user whose acoustic profile and emotional response patterns have never been observed during training. Random splitting of frames or windows, the default in most generic machine learning pipelines, would place frames from the same participant into both the training and test partitions, leading to optimistic performance estimates that do not reflect the difficulty of the cross-subject generalisation task. To avoid this, all splits in this work are constructed at the level of participant identity using scikit-learn’s GroupShuffleSplit utility, which guarantees that no participant appears in more than one partition.

The split proportions are 60% for training, 20% for validation, and 20% for testing, realised through two consecutive GroupShuffleSplit applications: a primary 60/40 split, followed by a 50/50 split of the 40% pool into validation and test. Both splits use a fixed random seed to ensure reproducibility. The validation partition serves a dual purpose in this work: during baseline training it acts as the early-

stopping criterion against which model checkpoints are selected, and during conformal calibration it acts as the calibration set from which the non-conformity scores are computed. The use of the validation partition for both purposes is a deliberate design choice, since the participants in this partition are entirely disjoint from those in the training set, so the non-conformity scores computed on it are not inflated by memorisation of training examples.

4.3.2 Feature Standardisation

The ComParE13 feature vectors in RECOLA and the engineered feature vectors in AGAIN have very different numerical ranges across dimensions, which can destabilise gradient-based optimisation and bias the loss towards the high-variance features. To address this, a scikit-learn StandardScaler is fitted on the training partition only, computing per-feature means and standard deviations, and is then applied identically to the validation and test partitions to produce zero-mean, unit-variance inputs. Fitting the scaler exclusively on the training data is essential to prevent statistical leakage of test-set information into the model, this principle applies to every transformation in the pipeline whose parameters are estimated from data.

4.3.3 Temporal Windowing

Both datasets provide affect annotations at frame-level resolution. For the static models (Linear and MLP), an individual frame contains no temporal context, for the LSTM, temporal structure across consecutive frames is precisely what the architecture exploits. To support all three architectures within a uniform input pipeline, the preprocessing module emits overlapping windows of $W = 20$ frames (0.8 s at 25 Hz), capturing local acoustic dynamics relevant to arousal. Each window is labelled by its final frame, consistent with next-step prediction, and windows do not span participant boundaries, preserving the participant-independent split structure.

4.3.4 Label Discretisation

To support the classification formulation of the affect recognition task, the continuous regression labels in both datasets are discretised into five ordinal classes. The class boundaries are derived from the 20th, 40th, 60th, and 80th percentiles of the training-set labels, with a small uniform jitter (on the order of 10^{-6}) added to break

ties at repeated label values. Because the boundaries are computed from the training labels only, the discretisation introduces no statistical leakage from the validation or test partitions.

Beyond the technical motivation of producing a multi-class formulation, label discretisation serves an important methodological purpose: it mitigates the subjectivity bias inherent in continuous emotion annotations. Two annotators may disagree on whether valence is 0.62 or 0.71 yet agree that the emotion is broadly pleasant. By collapsing the continuous scale into a small number of ordinal bins, the classification formulation absorbs fine-grained inter-annotator disagreement while preserving the coarse-grained affective signal that annotators share.

The choice of five classes, rather than two or three, is a compromise between informativeness and class balance. Quantile-based binning produces classes that are balanced by construction on the training set, so a uniform-random baseline achieves an expected 20% accuracy. This is small enough to leave substantial headroom for the trained classifiers, yet large enough that the prediction sets produced by the conformal classifier carry meaningful information at moderate confidence levels.

4.4 Baseline Model Design

Three architectures of increasing modelling capacity serve as baselines in this dissertation. The choice of architectures is deliberate: rather than aiming for state-of-the-art performance, the goal is to span a sufficiently wide range of model complexity that the interaction between model accuracy and conformal efficiency can be observed directly. All three architectures share the same input tensor shape (batch, W , F) and the same output head dimensionality, which is a scalar for regression tasks and a five-dimensional logit vector for classification tasks.

4.4.1 Linear Baseline

The Linear baseline applies a single fully-connected layer to the final frame of the input window, that is, to the slice $x[:, -1, :]$ of the input tensor. For regression, this corresponds to ordinary linear regression on the most recent feature frame, for classification, it corresponds to a multinomial logistic regression model with five output logits. The Linear baseline serves as a sanity check that the chosen features

carry a discriminative signal at all, and provides a lower bound on the predictive power that any more sophisticated architecture must surpass to justify its additional complexity.

4.4.2 Multi-Layer Perceptron

The Multi-Layer Perceptron (MLP) baseline applies a feedforward network with two hidden layers, of dimensionality 128 and 64 respectively, each followed by a ReLU activation and a dropout layer (with dropout probabilities $p = 0.3$ and $p = 0.2$). Like the Linear baseline, the MLP operates on the final frame of the input window, $x[:, -1, :]$, its capacity advantage over the Linear baseline is therefore confined to the non-linearity of the mapping from frame to output, not to any modelling of temporal context. Comparing the MLP to the Linear baseline isolates the contribution of model non-linearity, while comparing the MLP to the LSTM isolates the contribution of temporal modelling.

4.4.3 Long Short-Term Memory

The Long Short-Term Memory (LSTM) baseline processes the full $W = 20$ frame sequence through a single recurrent layer with a 64-dimensional hidden state, followed by a fully-connected output head applied to the final hidden state, $\text{out}[:, -1, :]$. The LSTM is the only one of the three architectures that has direct access to the temporal ordering of the feature frames within the window, any improvement in CCC or classification accuracy over the MLP can therefore be attributed to the modelling of temporal dynamics rather than to additional non-linear capacity, since the LSTM's parameter count is comparable to that of the MLP.

4.4.4 Common Output Head

All three architectures share the same output convention: a single linear projection from the penultimate representation to either a scalar (for regression) or a five-dimensional logit vector (for classification). For regression, the output is interpreted directly as the predicted affect value, for classification, the logits are passed through a softmax to obtain class probabilities, which are required by the hinge non-conformity score described in Section 4.5. Mean squared error is used as the regression loss and cross-entropy as the classification loss, both of which align

with the standard maximum-likelihood interpretation of the respective output heads under Gaussian and multinomial likelihoods.

4.5 Conformal Prediction Design

Each trained baseline model is wrapped with an Inductive Conformal Predictor using the crepes library [10]. The choice of crepes over alternative libraries such as MAPIE was discussed in Section 3.2, the key advantages are its explicit decoupling of the non-conformity score computation from the calibration and prediction steps, and its native support for both regression intervals and classification prediction sets. The conformal design is described separately for the regression and classification tasks below.

4.5.1 Conformal Regression

For each regression task, the trained baseline model is applied to the calibration partition to produce point predictions. The per-example absolute residual non-conformity score is defined as:

$$s_i = |y_i - \hat{y}_i| \quad (4.1)$$

The $(1 - \alpha)$ quantile of these calibration scores defines the half-width of the prediction interval at significance level α .

4.5.2 Conformal Classification

For each classification task, the trained classifier is applied to the calibration partition to produce softmax probability vectors over the five ordinal classes. The hinge non-conformity score is defined as:

$$s_i = 1 - \tilde{p}_{y_i} \quad (4.2)$$

where $\tilde{p}_{y_i}$ denotes the softmax probability assigned to the true class y_i . A prediction set at level α includes all classes whose softmax probability exceeds 1 minus the $(1 - \alpha)$ calibration quantile.

4.5.3 Significance Levels

Conformal coverage and efficiency are evaluated at five significance levels, $\alpha \in \{0.01, 0.05, 0.10, 0.20, 0.40\}$, corresponding to target confidence levels of 99%, 95%,

90%, 80%, and 60%. This range was chosen to expose the behaviour of the conformal predictor at both ends of the operating spectrum: at low α , the intervals and sets are wide and the empirical coverage is expected to be saturated near 100%, providing a test of the upper-tail behaviour of the calibration distribution, at high α , the intervals and sets are tight and the empirical coverage approaches the lower end of the useful operating range, providing a stringent test of the calibration in regions where many predictions are excluded from the set.

4.6 Evaluation Protocol

The evaluation protocol distinguishes carefully between the predictive performance of the baseline models on the one hand and the validity and efficiency of the conformal wrappers on the other. The two are reported separately because they answer different questions: the baseline metrics quantify how accurate the underlying point predictor is, while the conformal metrics quantify the quality of the uncertainty estimate that the wrapper produces around that point prediction.

For regression tasks, baseline predictive performance is reported using both the Concordance Correlation Coefficient (CCC, Equation 2.1) and the Root Mean Squared Error (RMSE). CCC is the standard metric in the AVEC benchmark series and penalises both correlation errors and scale or bias errors, RMSE is reported as a complementary metric on the same scale as the labels themselves, which is useful for interpreting the absolute magnitude of the prediction errors. For classification tasks, baseline predictive performance is reported using top-1 accuracy on the held-out test set, with the 20% uniform-random level serving as the natural reference baseline given the quantile-based class construction.

For the conformal evaluation, two metrics are computed at each of the five significance levels. The first is empirical coverage, defined as the fraction of test examples for which the true label lies within the predicted interval (regression) or set (classification). The marginal coverage guarantee requires that empirical coverage be at least $1 - \alpha$ in expectation, observed values above this level are valid but possibly conservative, while observed values significantly below this level constitute evidence that the exchangeability assumption is not satisfied for that experimental configuration. The second metric is efficiency: the Mean Prediction Interval Width

(MPIW) for regression and the Mean Prediction Set Size (MPSS) for classification. Smaller values of MPIW and MPSS at a fixed coverage level indicate a more informative predictor, the trade-off between coverage and efficiency is examined explicitly across the five significance levels.

All metrics are computed on the participant-independent test partition, which has not been seen by either the baseline model during training or the conformal wrapper during calibration. This ensures that the reported numbers reflect genuine generalisation to new users rather than within-subject memorisation, and provides the most realistic estimate of the predictor’s behaviour in deployment. Finally, it must be emphasised that no hyperparameter tuning was performed for any baseline architecture beyond the defaults specified in Section 4.4. The objective is to investigate the conformal framework on standard, untuned baselines rather than to maximise predictive performance, tuning on calibration data would risk conflating improvements in conformal metrics with improvements in the underlying point predictor.

5 Implementation

This chapter describes the implementation of the experimental pipeline. Five Python modules, summarised in Table 1, realise the design decisions of Chapter 4 as reproducible code. Each section gives the design intent for one module, library-level details are documented in the source comments.

Table 1: Summary of implementation modules

Module	Script	Responsibility
Data Inspection	inspect_data.py	Non-destructive audit: file counts, feature dimensionalities, label ranges, participant inventory
Preprocessing	process_data.py	Participant-independent split, feature standardisation, temporal windowing, label discretisation, .npy serialisation
Baseline Models	models.py	Three torch.nn.Module architectures (Linear, MLP, LSTM) with a uniform constructor interface
Training	train.py	18-configuration experiment grid, Adam optimiser, checkpoint serialisation
Conformal Prediction	conformal.py	ICP wrappers (regression and classification) using crepes, evaluation at five significance levels

5.1 Data Inspection

The `inspect_data.py` script performs a non-destructive audit of both datasets before any modelling, checking file counts, feature dimensionalities, label ranges, participant counts, and temporal window availability. Its outputs informed the preprocessing design choices documented in Section 4.3.

5.2 Preprocessing Implementation

The `process_data.py` module operationalises the four-step preprocessing design of Section 4.3. A shared `split_and_save` function applies `GroupShuffleSplit` with seed 42 to enforce the 60/20/20 participant-independent partition, fits a `StandardScaler` on training data only, applies 20-frame temporal windowing, performs quantile-based label discretisation from training labels, and serialises all partitions as NumPy `.npy` files under `DataSetsProcessed/` with a consistent `{dataset}_{X|y}_{target}_{task}_{split}.npy` naming convention.

5.3 Model Implementation

The `models.py` module defines the three baseline architectures as subclasses of `torch.nn.Module`, each accepting `input_dim` and `output_dim` at construction. The uniform constructor signature allows the training loop to instantiate any architecture interchangeably. Regression tasks use `output_dim = 1`, classification tasks use `output_dim = 5`. All output activations are deferred to the loss function.

5.4 Training Implementation

The `train.py` script constructs an 18-configuration experiment grid (3 architectures x 6 task variants) and trains each configuration using the Adam optimiser (`lr = 0.001`, `batch size = 64`, 50 epochs). Trained model checkpoints are saved to disk and consumed by the conformal evaluation script in Section 5.5.

5.5 Conformal Implementation

The `conformal.py` script loads each trained baseline checkpoint and wraps it with an Inductive Conformal Predictor from the `crepes` library [10]. For regression, it fits a `ConformalRegressor` on the validation partition using absolute residuals (Equation 4.1) and evaluates prediction intervals on the test set. For classification, it fits a `ConformalClassifier` using hinge non-conformity scores (Equation 4.2) and evaluates prediction sets. Coverage and efficiency metrics are computed at all five significance levels and logged for analysis in Chapter 6.

6 Testing and Evaluation

This chapter reports empirical results for the eighteen experimental configurations defined in Chapter 4. All numbers are computed on the participant-independent test partition that was unseen by both the trained baselines and the conformal calibration step.

6.1 Baseline Model Performance

Baseline regression and classification performance on the held-out test partition is reported in Table 2. Three patterns are immediately visible. First, on RECOLA arousal regression the LSTM dramatically outperforms both static baselines, achieving $CCC = 0.81$ against approximately 0.42 for the Linear and MLP models. This factor-of-1.9 gap, with a corresponding 36% reduction in RMSE, indicates that temporal context across the 20-frame (0.8-second) window carries substantial information for arousal prediction from speech audio, consistent with the AVEC-series baselines reported by Makantasis et al. [3].

Second, on AGAIN arousal regression all three architectures perform almost identically ($CCC \approx 0.42$), with no meaningful benefit from the recurrent architecture. This is consistent with two factors documented in the literature: the high inter-participant variability identified by Melhart et al. [2] as a defining property of gameplay arousal, and the absence of a strong short-horizon temporal signal in self-reported arousal that the LSTM is in a position to exploit. Interestingly, the LSTM has the highest RMSE on AGAIN (0.264 against 0.248 for Linear), an empirical observation that has direct consequences for conformal efficiency in Section 6.2. Third, valence prediction is uniformly harder than arousal on RECOLA. The best valence regressor (LSTM) achieves $CCC = 0.42$, half its arousal counterpart, in line with the long-standing finding that valence is poorly recoverable from audio alone [17]. The classification view confirms this pattern: RECOLA valence classification accuracy peaks at 32.5% (LSTM), only 12.5 percentage points above the 20% random baseline, whereas RECOLA arousal classification reaches 51.3%, more than double the random baseline.

Table 2: Baseline model performance

Configuration	Linear	MLP	LSTM
RECOLA arousal regression (CCC / RMSE)	0.425 / 0.189	0.425 / 0.180	0.814 / 0.121
RECOLA valence regression (CCC / RMSE)	0.180 / 0.139	0.216 / 0.137	0.422 / 0.131
AGAIN arousal regression (CCC / RMSE)	0.422 / 0.248	0.429 / 0.255	0.421 / 0.264
RECOLA arousal classification (accuracy)	0.393	0.412	0.513
RECOLA valence classification (accuracy)	0.249	0.272	0.325
AGAIN arousal classification (accuracy)	0.342	0.348	0.336

6.2 Conformal Regression Results

Conformal regression results are summarised in Table 3, which reports empirical coverage (%) and the Mean Prediction Interval Width (MPIW) at each of the five significance levels, and comparison of empirical and target coverage is depicted in Figure 4. On RECOLA, empirical coverage tracks the target confidence level closely across all five significance levels and all three models. The largest absolute deviation in the RECOLA rows is 16.1 percentage points (MLP valence at $\alpha = 0.40$, which achieves 76.1% empirical coverage against a 60% target). However, all such deviations are overshoots, the predictor is conservative rather than deficient, and no RECOLA configuration falls below its nominal coverage level.

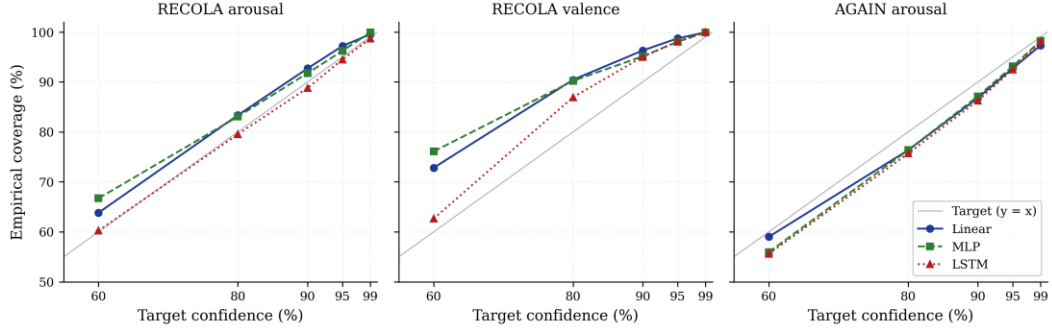


Figure 4: Empirical versus target coverage for conformal regression

On AGAIN, by contrast, coverage falls consistently below target at every significance level and for all three architectures (Figure 5). Averaged across models, the shortfall is 1.1 percentage points at the 99% target, widens to 2.2 points at 95%, reaches 3.3 points at 90% and 3.9 points at 80%, and is 3.2 points at 60%. The monotone broadening of this gap with α and its persistence across architectures is the empirical signature of a systematic exchangeability violation rather than sampling fluctuation: AGAIN sessions are continuous gameplay traces with strong intra-session temporal autocorrelation, so the calibration residuals computed on validation participants are not exchangeable with the test residuals at the frame level. To the author’s knowledge, this is the first quantitative report of conformal coverage behaviour on high-frequency gaming arousal data.

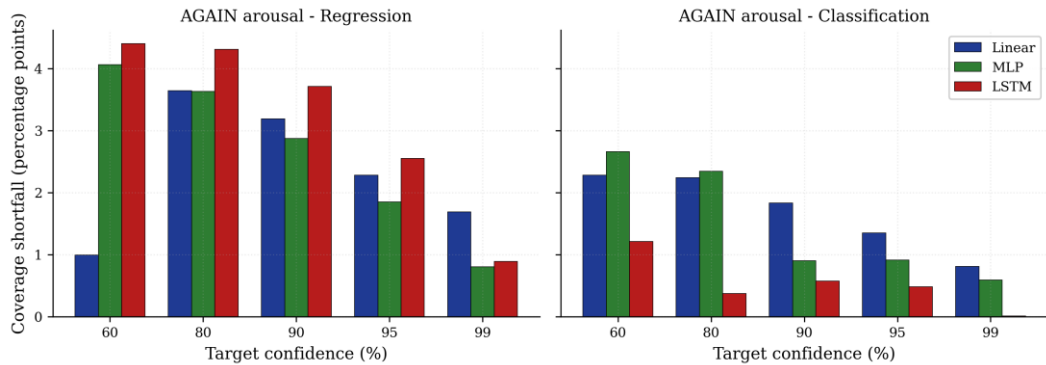


Figure 5: Coverage shortfall on AGAIN arousal regression

In efficiency terms, the LSTM produces substantially narrower intervals than either static baseline on RECOLA arousal, 0.409 at $\alpha = 0.10$ against 0.578–0.614 for

the static models, a 33% reduction, directly mirroring its higher CCC. On RECOLA valence the three models yield comparable widths because their accuracy differences are small. On AGAIN, the LSTM produces marginally wider intervals than the static baselines (0.853 against 0.793 at $\alpha = 0.10$), an artefact of its slightly higher RMSE on this dataset. Within every task variant the ordering of MPIW values matches the ordering of RMSE values exactly, empirically confirming the efficiency–accuracy link emphasised by Molnar [9].

Table 3: Conformal regression (*empirical coverage (%) / MPIW*)

Task	Model	$\alpha=0.01$ (99%)	$\alpha=0.05$ (95%)	$\alpha=0.10$ (90%)	$\alpha=0.20$ (80%)	$\alpha=0.40$ (60%)
RECOLA arousal	Linear	99.5 / 0.890	97.2 / 0.726	92.7 / 0.614	83.4 / 0.491	63.8 / 0.340
	MLP	99.9 / 0.877	96.2 / 0.681	91.8 / 0.578	83.1 / 0.465	66.7 / 0.322
	LSTM	98.7 / 0.632	94.5 / 0.493	88.8 / 0.409	79.5 / 0.310	60.2 / 0.188
RECOLA valence	Linear	99.9 / 0.755	98.7 / 0.571	96.3 / 0.461	90.4 / 0.348	72.8 / 0.222
	MLP	99.9 / 0.777	98.0 / 0.566	95.1 / 0.445	90.2 / 0.326	76.1 / 0.217
	LSTM	100.0 / 0.821	98.1 / 0.533	95.0 / 0.423	86.9 / 0.314	62.7 / 0.193
AGAIN arousal	Linear	97.3 / 1.219	92.7 / 0.933	86.8 / 0.793	76.4 / 0.629	59.0 / 0.448
	MLP	98.2 / 1.316	93.2 / 0.977	87.1 / 0.824	76.4 / 0.651	55.9 / 0.442
	LSTM	98.1 / 1.420	92.5 / 1.021	86.3 / 0.853	75.7 / 0.672	55.6 / 0.446

6.3 Conformal Classification Results

Conformal classification results appear in Table 4, which reports empirical coverage (%) and Mean Prediction Set Size (MPSS) per cell, while Figure 6 illustrates their graph relation. The coverage pattern broadly parallels the regression case but the AGAIN shortfall is markedly smaller: averaged across models, the AGAIN classification shortfall is 0.5 points at 99%, 0.9 at 95%, 1.1 at 90%, 1.7 at 80%, and 2.1 at 60%, less than half the regression shortfall at every level. The likely mechanism is that the hinge non-conformity score is bounded in $[0, 1]$ and quantises model output through five discrete softmax peaks, both of which compress the effect of temporal autocorrelation on the calibration distribution. On RECOLA the LSTM classification coverage is within ± 3 points of every target, the static models on RECOLA arousal classification show somewhat larger undershoots at moderate-to-high α (e.g. 87.1% at the 90% target, 73.4–75.6% at the 80% target), consistent with their flatter softmax outputs producing a less informative hinge distribution.

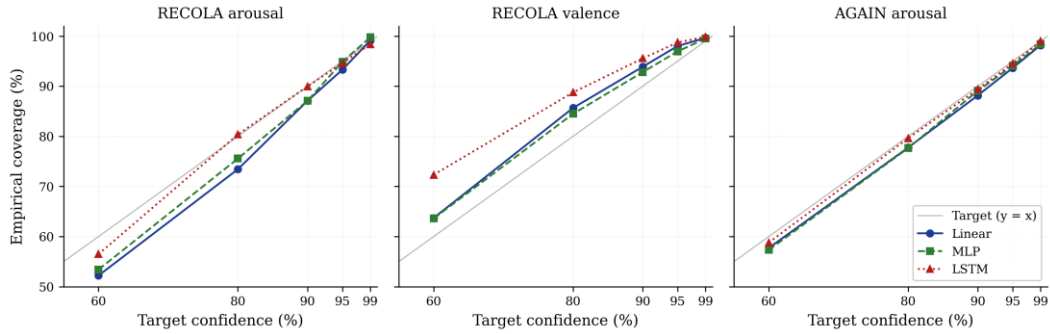


Figure 6: Empirical versus target coverage for conformal classification

The set-size results are the most diagnostically useful output of the entire pipeline. For RECOLA arousal at $\alpha = 0.10$ the LSTM produces an average prediction set of 2.94 classes, roughly halving the five-class hypothesis space, and at $\alpha = 0.40$ the set shrinks to 1.33 classes, meaning that for the majority of test windows the conformal classifier confidently selects a single class while still satisfying the requested coverage. By contrast, the same predictor on RECOLA valence yields 4.30 classes at $\alpha = 0.10$, barely more informative than a uniform set. This is direct

quantitative confirmation that conformal sets faithfully report the residual uncertainty in tasks where the underlying signal is weak, precisely as the framework is designed to do. On AGAIN, set sizes are intermediate (3.78–3.91 at $\alpha = 0.10$) and broadly uniform across architectures, reflecting the comparable underlying accuracies ($\approx 34\%$) of the three classifiers.

Table 4: Conformal classification

Task	Model	$\alpha=0.01$ (99%)	$\alpha=0.05$ (95%)	$\alpha=0.10$ (90%)	$\alpha=0.20$ (80%)	$\alpha=0.40$ (60%)
RECOLA arousal	Linear	99.1 / 4.81	93.3 / 4.24	87.1 / 3.81	73.4 / 3.02	52.2 / 1.93
	MLP	99.7 / 4.82	94.9 / 4.17	87.1 / 3.67	75.6 / 3.00	53.4 / 1.86
	LSTM	98.4 / 4.24	94.6 / 3.49	90.0 / 2.94	80.4 / 2.32	56.5 / 1.33
RECOLA valence	Linear	99.6 / 4.90	98.0 / 4.72	93.9 / 4.38	85.7 / 3.81	63.7 / 2.67
	MLP	99.5 / 4.92	96.9 / 4.64	92.8 / 4.32	84.5 / 3.73	63.6 / 2.62
	LSTM	99.9 / 4.92	98.7 / 4.63	95.6 / 4.30	88.8 / 3.64	72.3 / 2.50
AGAIN arousal	Linear	98.2 / 4.75	93.7 / 4.23	88.2 / 3.78	77.8 / 3.12	57.7 / 2.11
	MLP	98.4 / 4.76	94.1 / 4.26	89.1 / 3.80	77.7 / 3.04	57.3 / 1.98
	LSTM	99.0 / 4.80	94.5 / 4.34	89.4 / 3.91	79.6 / 3.25	58.8 / 2.14

6.4 Discussion

The five objectives stated in Section 1.3 can now be assessed. Objectives 1 and 4 (data acquisition and participant-independent evaluation) are evidenced by every reported number, since no participant appears in more than one partition. Objective 2 (baseline development) is evidenced by Table 2, where all three architectures produce above-chance performance across every task variant. The remainder of this section addresses Objectives 3 and 5.

The central finding is that conformal prediction delivers its theoretical coverage guarantee on RECOLA across both arousal and valence and across both regression and classification, with no modification to the trained baselines. Mean absolute deviations from target are generally below three percentage points, consistent with finite-sample fluctuation, validating ICP as an off-the-shelf uncertainty-quantification tool for laboratory-recorded affect data.

On AGAIN, the systematic two-to-four percentage-point coverage shortfall in regression, present at every confidence level and across every architecture, constitutes the most consequential empirical contribution of this dissertation. It demonstrates that the exchangeability assumption is not innocuous for high-frequency gameplay data. The shortfall is small enough that the framework remains practically useful (a 90% nominal interval still attains approximately 87% empirical coverage), but it must be reported honestly and motivates the future work in Section 7.1.

A second key finding is the strict efficiency–accuracy relationship: across all six task variants, every reduction in baseline RMSE translates monotonically into a narrower MPIW at fixed coverage. The LSTM on RECOLA arousal regression most clearly illustrates this, producing intervals 33% narrower than the static baselines at $\alpha = 0.10$. The practical implication is that any improvement to the underlying point predictor propagates immediately into tighter prediction intervals without any change to the conformal procedure.

A third, less expected, observation is that conformal classification is empirically more robust than conformal regression to the AGAIN exchangeability violation, with a classification shortfall less than half the regression shortfall at every confidence level. The likely mechanism is that the bounded hinge score and discrete five-class output absorb part of the temporal-correlation effect that an unbounded residual propagates into the calibration distribution, suggesting that the classification formulation may be the more conservative choice when exchangeability is in doubt.

Taken together, these results substantiate the central claim of the dissertation: conformal prediction is a viable, computationally lightweight, and statistically grounded approach to uncertainty quantification in affective computing, with concrete practical benefits and clearly characterised failure modes.

7 Future Work

The main findings of Chapter 6 and the methodological limitations acknowledged in Section 6.4 motivate four concrete directions for future work. Addressing the observed coverage shortfall on AGAIN, strengthening the coverage guarantee from marginal to conditional, improving the underlying point predictors, and extending the framework to real-time deployment.

7.1 Addressing the Exchangeability Violation in AGAIN

The coverage shortfall on AGAIN regression, a systematic two-to-four percentage-point undershoot at every confidence level, is the most pressing open problem identified by this work. Its root cause is the temporal autocorrelation of self-report annotations, which violates the exchangeability assumption of standard ICP. Two remedies deserve investigation: block-based calibration strategies, which treat contiguous windows of non-conformity scores as a single exchangeable block (analogously to block bootstrapping), and Adaptive Conformal Inference [23], which reformulates calibration as an online problem and naturally accommodates non-stationary label dynamics within a gaming session.

7.2 Conditional Coverage

The Inductive Conformal Predictor evaluated in this dissertation provides marginal coverage: the guarantee $P(Y \in \Gamma) \geq 1 - \alpha$ holds in expectation over the joint distribution of inputs and labels, but not necessarily for any specific subgroup of the test population. In an affective-computing context this is a substantive limitation. A system that delivers 90% coverage on average but only 60% coverage on, for instance, participants with atypical vocal characteristics, or on moments of low arousal, may still be unsafe to deploy even though its aggregate behaviour appears valid.

7.3 Stronger Baseline Architectures

The baseline architectures used in this dissertation were chosen deliberately for their simplicity and standard interpretability, not for state-of-the-art predictive performance. Because conformal prediction is model-agnostic, the framework established in Chapters 4 and 5 can be re-applied without modification to any improved point predictor, and the efficiency-accuracy relationship demonstrated in Section 6.4 ensures that gains in underlying model accuracy translate directly into tighter prediction intervals or smaller prediction sets.

Two directions are particularly promising: replacing the LSTM baseline with a Transformer-based audio encoder (pre-trained or task-specific), and incorporating a conformally calibrated wav2vec 2.0 or HuBERT feature extractor. Because conformal prediction wraps any trained model without modification, stronger baselines would directly and proportionally narrow prediction intervals, as the efficiency-accuracy relationship established in Section 6.4 predicts.

7.4 Real-Time Deployment

This dissertation evaluates conformal predictors in an offline setting in which both the calibration set and the test set are fixed and available in their entirety. The most impactful practical application of the framework, however, is in real-time interactive systems that act on the uncertainty estimate as it is produced. The motivating example is an adaptive game that increases or decreases its challenge level based on the player’s predicted arousal.

Deployment introduces two technical challenges. First, online calibration is needed: as a participant generates new affect data, the calibration set must grow incrementally to maintain coverage validity without retraining the base model. Second, the exchangeability assumption must be addressed under distributional shift within and across sessions, for instance through the Adaptive Conformal Inference approach of [23]. Resolving these challenges would make conformal affect recognition deployable in adaptive game AI, mental-health monitoring, and affective human-computer interaction.

8 Conclusions

This dissertation investigated whether Conformal Prediction can provide statistically valid coverage guarantees for affective computing models without modifying their architecture or training procedure. The empirical evidence, gathered across two affective datasets, three baseline architectures, and eighteen experimental configurations, supports an affirmative answer for laboratory-recorded spontaneous affect, qualified by a clearly characterised shortfall in the temporally autocorrelated gaming setting. This chapter summarises the contributions of the work in relation to the five objectives, distils the principal empirical findings, acknowledges the limitations, and closes with brief remarks on the practical value of the framework for the affect modelling community.

8.1 Summary of Contributions

All five objectives defined in Section 1.3 were fulfilled. A reproducible, participant-independent preprocessing pipeline was implemented for both RECOLA and AGAIN, eliminating data leakage at the feature-standardisation and label-discretisation steps and yielding the eighteen-configuration experimental matrix described in Chapter 4. Three baseline architectures, a Linear model, a Multi-Layer Perceptron, and an LSTM, were trained and evaluated across regression and classification formulations of arousal and valence prediction, providing a controlled comparison spanning a range of modelling capacity. Conformal Predictors were then calibrated on held-out validation participants using the crepes library [10] and evaluated at five significance levels covering target confidence levels from 99% to 60%.

In addition to these technical deliverables, the dissertation contributes a comparative evaluation framework that has not, to the author’s knowledge, been applied previously to either RECOLA arousal and valence regression or AGAIN gameplay arousal under participant-independent splits. The full code base, intermediate artefacts, and evaluation logs are released as an open-access GitHub

repository¹ accompanying this dissertation, addressing the open-source deliverable objective stated in Section 1.3 and providing a foundation on which the future work directions identified in Chapter 7 can be built.

8.2 Key Findings

Three empirical findings emerge from the evaluation reported in Chapter 6. The first, and most important, is that Inductive Conformal Prediction reliably delivers its theoretical coverage guarantee on RECOLA for both arousal and valence and across both regression and classification tasks, with mean absolute deviations from the nominal rate generally below three percentage points. This establishes CP as a viable, computationally lightweight, and formally grounded approach to uncertainty quantification in laboratory-recorded affective data, with the practical advantages over Bayesian Neural Networks, Monte Carlo Dropout, and Deep Ensembles that the literature review of Chapter 3 identified.

The second finding is a systematic two-to-four percentage-point coverage shortfall on AGAIN regression, persistent across architectures and significance levels. This is both a limitation of vanilla ICP under temporally autocorrelated data and, in itself, a quantitative empirical contribution: it characterises the conditions under which the exchangeability assumption is strained in high-frequency interactive affect, an observation that to the author’s knowledge has not been reported previously in the conformal prediction literature on affective computing. The shortfall is small enough that the framework remains practically useful, a 90% nominal interval still attains roughly 87% empirical coverage, and motivates the targeted methodological extensions discussed in Section 7.1.

The third finding is the strict monotonic relationship between baseline accuracy and conformal efficiency. An auxiliary observation, suggested by the present data and worth further investigation, is that conformal classification appears empirically more robust than conformal regression to the AGAIN exchangeability violation, plausibly because the bounded hinge non-conformity score and the discrete

¹ The full source code, data artefacts, and evaluation logs are publicly available at: <https://github.com/romvasil/FYP>

five-class output absorb part of the temporal-correlation effect that an unbounded residual propagates into the calibration distribution.

8.3 Limitations

Four principal limitations bound the scope of the present work. First, the baseline architectures were deliberately kept simple, prioritising a clean attribution of conformal evaluation outcomes to the calibration procedure over absolute predictive performance, no hyperparameter tuning was performed and no state-of-the-art comparison was attempted. Stronger architectures, such as the Transformer-based and multimodal-fusion variants identified in Section 7.3, are likely to produce both higher baseline accuracy and proportionally tighter conformal predictions.

Second, the AGAIN analysis was restricted to the Shooter genre to control for inter-genre confounds, whether the observed exchangeability violation manifests with comparable magnitude in the Platformer and Racing genres remains an open empirical question.

Third, all coverage guarantees reported in this dissertation are marginal: validity is established in aggregate across the test population, but not within subgroups defined by participant, intensity, or session phase, as discussed in Section 7.2. Fourth, all evaluation was conducted offline, the framework has not been tested in a live interactive setting with the latency, distribution shift, and online calibration challenges that real-time deployment would introduce, as discussed in Section 7.4.

8.4 Closing Remarks

A model that cannot communicate uncertainty is a model that cannot be trusted. The central message of this dissertation is that the long-standing tension between the subjective, distributional nature of human emotion and the deterministic, point-estimate outputs of standard machine learning models can be resolved, at least partially, by a framework that is principled in its guarantees, agnostic to its underlying model, and cheap to deploy. Conformal prediction provides exactly such a framework, and the empirical work reported here demonstrates that it is ready for adoption on laboratory-controlled affect corpora with no modification, while requiring targeted methodological extensions for high-frequency interactive contexts.

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